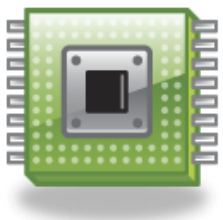


Homework 2 (Deadline : 2025/10/28 Tue. 23:59)

- Please refer to the following two pages for the assignment details. There are two RTL circuit design problems, and both must be solved.

- It‘s important to note that these two circuit problems must be solved using only combinational building blocks such as adders, comparators, multiplexers, decoders, encoders, and arbiters, as well as logic gates. The use of flip-flops, counters, or other sequential circuits is not permitted.



Homework 2 (Deadline : 2025/10/28 Tue. 23:59)

1. Vote Counting and Reporting Circuit Design: Design a circuit that accepts five three-bit one-hot (i.e., only one bit will have the logic-1 value) inputs and produces two outputs:
   1) The three-bit one-hot number that appeared most frequently among the inputs.
   2) A 3-bit binary count indicating how many times that number appeared.

Ties can be broken arbitrarily. For example, if the inputs are 100, 100, 010, 010, and 001, the output will be 100.

Homework 2 (Deadline : 2025/10/28 Tue. 23:59)

2. **Median circuit design**: Design a circuit that accepts **three eight-bit one-hot numbers**, $a2_{7:0}$, $a1_{7:0}$, $a0_{7:0}$, and **outputs the input with the middle of the three values**. For example, if the inputs a2=10000000, a1= 00010000, and a0= 00000001, the output should be 00010000, the median of the three one-hot values.